

Efficacy, safety, and biomarker changes of B-cell activating factor- and A proliferation-inducing ligand-targeted therapies in IgA nephropathy: A systematic review and meta-analysis of randomized controlled trials

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Abstract

Background: IgA nephropathy (IgAN) is the most common primary glomerular disease worldwide. BAFF (B-cell activating factor) and APRIL (A proliferation-inducing ligand) are cytokines involved in B-cell activation and survival, contributing to the pathogenesis of IgAN. This meta-analysis aimed to evaluate the efficacy, safety, and biomarkers of BAFF and APRIL-targeted therapies in patients with IgAN.

Methods: This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and was registered with PROSPERO (CRD42024598157). We searched PubMed, Scopus, and the Cochrane Library for randomized controlled trials (RCTs) comparing BAFF or APRIL-targeted drugs with placebo in adults with biopsy-confirmed IgAN. The efficacy outcomes were the mean percent change in the urine protein to creatinine ratio (UPCR) at 24 hours and the mean change in the estimated glomerular filtration rate (eGFR) from baseline. Safety included the incidence of adverse events. The biomarkers were changes in serum Gd-IgA1, IgG, IgA, and IgM from baseline. We used the R software version 4.2.1 for statistics.

Results: Four phase II RCTs including 331 patients were included. Compared to placebo, treatment significantly reduced 24-hour UPCR (mean difference [MD] -38.94% ; 95% confidence interval [CI] -58.98 to -18.90 ; $p = 0.0001$; $I^2 = 0\%$) and significantly improved eGFR (MD 7.05 mL/min/ 1.73 m 2 ; 95% CI 3.83 to 10.27 ; $p < 0.0001$; $I^2 = 0\%$). The incidence of adverse events did not differ significantly in the treatment and placebo groups. The drug significantly decreased serum Gd-IgA1, IgG, IgA, and IgM compared to placebo.

Conclusions BAFF and APRIL-targeted therapies effectively and safely reduce proteinuria in patients with IgAN.

INTRODUCTION

IgA nephropathy (IgAN) is the most common primary glomerular disease in the world. IgAN is characterized by the deposition of abnormally glycosylated IgA1 (Gd-IgA1) immune complexes in the glomerular mesangial matrix, which promotes inflammation and progressive renal dysfunction [1]. Despite the great variability in clinical presentation, many patients develop a persistent decline in renal function, progressing to end-stage renal disease (ESRD) in 10%-60% of cases within 10 years and up to 40% within 20 years [2].

The currently available therapy for IgAN focuses on the management of symptoms by reducing proteinuria and controlling blood pressure with angiotensin converting enzyme inhibitors (ACEIs) and angiotensin receptor blockers (ARBs), as recommended by international guidelines [3]. However, these strategies are often insufficient to slow the progression of the disease, especially in patients with persistent proteinuria [2].

Advances in understanding IgAN pathophysiology have led to the exploration of new therapeutic strategies targeting disease mechanisms. BAFF (B-cell activating factor) and APRIL ((A proliferation-inducing ligand) are cytokines that play central roles in the activation and survival of specific B-cell populations responsible for IgA production [4]. These cytokines are directly involved in the production of Gd-IgA1 and the formation of immune complexes, and their circulation levels have been directly correlated with both the clinical and histological severity of IgAN [5]. Unlike conventional therapies that aim primarily to reduce proteinuria and control blood pressure, BAFF and APRIL-targeted therapies act on the immunological root of the disease by modulating B-cell responses and reducing Gd-IgA1 production [6]. In this context, BAFF/APRIL inhibitors, including atacicept, sibeprenlimab, and telitacicept, have been investigated in clinical studies as adjunctive agents for patients with IgAN, showing the ability to reduce proteinuria, reduce disease severity, stabilize kidney function, improve serum biomarkers, and slow disease progression in patients who failed conventional therapy [5–9].

However, the restricted number of clinical trials, which are all Phase II with relatively small sample sizes [5–8], requires a thorough data synthesis to fully evaluate the efficacy, safety and biomarker changes of BAFF and APRIL therapy in patients with IgAN. Therefore, we performed a systematic review and meta-analysis in order to refine the understanding of individual results in terms of efficacy, safety, and biomarker changes.

MATERIALS AND METHODS

The present systematic review and meta-analysis is registered in the International Prospective Register of Systematic Reviews (PROSPERO) [10] under the registration number CRD42024598157. This systematic review with meta-analysis was carried out following the methodological principles delineated in the Cochrane Handbook for Systematic Reviews of Interventions [11], and the findings are reported following the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines [12].

Eligibility criteria

The inclusion criteria for this systematic review with meta-analysis were: (1) randomized controlled trials; (2) that directly compared BAFF or APRIL-targeted drugs versus placebo; (3) in adult patients aged ≥ 18 years old; (4) with biopsy-confirmed IgA nephropathy; (5) published in the English language. On the contrary, we excluded studies that were nonrandomized controlled trials, case reports or series, in vitro studies, animal studies, reviews, book chapters, opinions, conference abstracts, overlapping populations, or those that lacked a comparative analysis of clinically relevant outcomes of BAFF or APRIL-targeted drugs.

Search strategy and data extraction

The databases PubMed, Scopus, and the Cochrane Library were searched without time limits until December 2024. Our search strategy used a combination of keywords, including: "IgA Nephropathy", "Berger disease", "Berger's disease", "Berger", "synpharyngitic glomerulonephritis", "synpharyngitic hematuria", "BAFF", "B-cell activation factor", "B-cell activating factors", "Povetacicept", "ALPN-303", "Telitacicept", "Atacicept", "Blisibimod", "APRIL", "A proliferation inducing ligand", "TNFSF13", "tumor necrosis factor ligand superfamily member 13", "TNF superfamily member 13", "Zigakibart", "BION-1301", "Sibeprenlimab", "VIS639", and "VIS649". The complete search strategy is presented in the supplementary information. Furthermore, we meticulously examined the reference lists of the included manuscripts to identify any additional relevant studies. Two authors (RDSB and LHDP) independently extracted the data following predefined search criteria. Any discrepancy was resolved by discussion in a consensus meeting between the authors.

Outcomes

The efficacy outcomes considered were the mean percent change in the urine protein to creatinine ratio (UPCR) at 24 hours from baseline and the mean change in the estimated glomerular filtration rate (eGFR) from baseline. Safety outcomes comprised the rate of adverse effects, drug- or placebo-related adverse effects, serious adverse effects, adverse events leading to discontinuation of the drug or placebo, and upper respiratory tract infections. The biomarker results were the mean percent change in galactose-deficient IgA1 (Gd-IgA1) from baseline; the mean percent change in IgG, IgA, and IgM from baseline; and the final levels of serum IgG, IgA, and IgM.

Risk of bias assessment

Two authors (RLS and RDSB) separately performed the risk of bias and quality assessment of the studies using the Cochrane Collaboration tool to assess the risk of bias in randomised studies (RoB v2.0) [13]. For each trial, the authors designated a score indicating 'high risk', 'low risk' or 'some concerns' in five domains: selection, performance, detection, attrition, and reporting bias. Publication bias was not evaluated using funnel plot asymmetry and Egger's test due to the low number of included studies, as recommended in the Cochrane Handbook for Systematic Reviews of Interventions [11]. The quality of evidence was evaluated using the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) approach, considering aspects that may increase or decrease the quality of evidence. The factors analyzed were study limitations, inconsistency, indirectness, imprecision, and publication bias. Ultimately, the final level of evidence was classified into high, moderate, low, and very low [14].

Statistical analysis

Risk ratios (RR) alongside 95% confidence intervals (CI) were calculated to compare the occurrence of binary endpoints between the two groups, while continuous results were evaluated as mean differences

(MD) with the corresponding 95% CI. For the conversion of continuous data initially reported in the median (interquartile range), we used the estimated mean and standard deviation provided by Luo et al. (2018) [15] and Wan et al. (2014) [16], respectively. The inverse variance (IV) method was used for continuous outcomes and the Mantel-Haenszel (MH) method for binary outcomes [11]. Cochran's Q Test and I^2 statistics were employed to assess heterogeneity. The endpoints were classified as having low heterogeneity if $p > 0.10$ and $I^2 < 30\%$, moderate heterogeneity if I^2 fell between 30 and 75%, and high heterogeneity if I^2 exceeded 75%. Significance was attributed to p values < 0.05 . We used the random effects model to account for potential variability among studies, which often yields a more realistic approach in real-world scenarios, in which study populations, interventions, and conditions differ, making it more suitable for generalizing results to different settings and populations. Furthermore, this approach covers unobserved sources of variability and can better handle heterogeneity if it arises in future updates of the meta-analysis as more studies are included [11]. Statistical analyses were performed using the packages 'meta' and 'metafor' on the R version 4.2.1 statistical software (R Foundation for Statistical Computing) [17].

Sensitivity analysis

The DerSimonian-Laird (DL) was used as the default method to estimate τ^2 [11]. However, we conducted a prespecified sensitivity analysis to evaluate the stability, robustness, and consistency of our findings, and to address any uncertainty or potential imprecision in the effect estimates. The data were re-assessed using the Sidik-Jonkman-Hartung-Knapp (SJHK) method, as recommended in the Cochrane Handbook (Section 10.10.4.4) for meta-analyses with a small number of studies or in cases where there is low apparent heterogeneity but potential uncertainty due to study sizes [11]. Additionally, a leave-one-out analysis was conducted by systematically eliminating one study at a time to ensure that the results were not disproportionately influenced by a single study [18].

RESULTS

Study selection and baseline characteristics

As shown in Fig. 1, the search strategy initially yielded 3,215 articles. After removing 1,175 duplicates, an additional 2,019 manuscripts were excluded based on title and abstract screening due to the lack of compliance with the research question. Subsequently, the remaining 21 studies were fully reviewed. Of these, 15 were excluded as abstracts from unpublished trials, 1 because it was a nonrandomized trial, and 17 for reporting preliminary results of a published trial. Therefore, four published RCTs met all the inclusion criteria. The non-overlapping population consisted of 331 patients with IgAN. The overall cohort was predominantly Asian (51.06%) and male (56.80%), with a mean age of 40.72 ± 26.74 years. Baseline urinary protein excretion was 3.46 ± 6.13 g/day, and baseline eGFR, estimated using the CKD-EPI equation, was 73.52 ± 60.79 mL/min/1.73 m². Table 1 provides an overview of the study characteristics and baseline demographics for the BAFF or APRIL-targeted drug and placebo groups.

Quality assessment

The four included RCTs followed a similar design, with ORIGIN, ENVISION, and Janus considered at low risk of bias in all RoB2 domains. The Jicheng study was flagged with some concerns due to deviations from the intended interventions (Fig. 2). Since we only included four RCTs in this meta-analysis, no formal assessment of publication bias using funnel plot asymmetry and Egger's test is recommended. Regarding the GRADE quality of the evidence (Table S1), the mean percentage change in 24-hour UPCR from baseline, the mean change in eGFR from baseline, the rate of drug- or placebo-related adverse events, the rate of serious adverse effects and the final level of serum IgG were considered with moderate certainty downgraded due to potential publication bias. The rate of any adverse effect and the mean percent change in Gd-IgA1 from baseline were also considered moderate certainty due to the heterogeneity of the pooled data and potential publication bias. The rate of adverse events leading to discontinuation of the study was rated as low certainty, downgraded due to the limited sample size and wide CI that may reduce confidence in the estimate of effect, and potential publication bias. Of particular interest, the rate of upper respiratory tract infection was considered with very low certainty due to (1) some concerns about the risk of bias assessment of the Jicheng Lv trial, which accounts for 74.24% of the weight of this outcome; (2) heterogeneity of the pooled data; (3) the limited sample size, which could result in unreliable statistical power to detect statistical significance; and (4) potential publication bias (Table S1).

Efficacy

The mean percent change in 24-hour UPCR from baseline showed a significant decrease with the use of BAFF or APRIL-targeted drugs compared to placebo (MD -38.80%; 95% CI -59.17 to -18.44; $p = 0.0002$; $I^2=0\%$; Fig. 3A). In the sensitivity analysis using the SJHK method, the superior effect of BAFF or APRIL-targeted drugs over placebo was maintained (MD -38.81%; 95% CI -59.21 to -18.40; $p = 0.0002$; $I^2=0\%$; Figure S1A). In the leave-one-out sensitivity analysis, the removal of each study from the pooled analysis did not affect this endpoint (Figure S1B).

The trial conducted by Jicheng Lv et al. (2023), which evaluated the drug Telitacicept, did not report UPCR. However, their data showed that Telitacicept 240 mg reduced the mean proteinuria 24 hours (g/day) by 49% from baseline (change in proteinuria vs. placebo 0.88; 95% CI -1.57 to -0.20; $p = 0.013$), whereas Telitacicept 160 mg reduced it by 25% (change in proteinuria vs. placebo - 0.29; 95% CI -0.95 to 0.37; $p = 0.389$) [8].

The mean change in eGFR from baseline showed a significant improvement with BAFF or APRIL-targeted drugs compared to placebo (MD 7.05 mL/min/1.73 m²; 95% CI 3.83 to 10.27; $p < 0.0001$; $I^2=0\%$; Fig. 3B). In the sensitivity analysis using the SJHK method, the superior effect of BAFF or APRIL-targeted drugs over placebo was maintained (MD 7.08 mL/min/1.73 m²; 95% CI 3.68 to 10.48; $p < 0.0001$; $I^2=0\%$; Figure S2A). In the leave-one-out sensitivity analysis, the removal of each study from the pooled analysis did not affect this endpoint (Figure S2B).

Safety

The rate of any adverse effect was comparable between the BAFF or APRIL-targeted drug and the placebo groups (BAFF or APRIL-targeted drug: 189/240 = 78.75%; placebo: 71/91 = 78.02%; RR 0.99; 95% CI 0.86 to 1.13; $p = 0.8381$; $I^2=25.9\%$; Fig. 4A). In the sensitivity analysis using the SJHK method, the outcome was consistently maintained (RR 0.98; 95% CI 0.85 to 1.15; $p = 0.8423$; $I^2=21.1\%$; Figure S3A). In the leave-one-out sensitivity analysis, the removal of each study from the pooled analysis did not affect this outcome, and the removal of the JANUS 2022 trial yielded homogeneity ($I^2=0\%$; Figure S3B).

Of particular interest, the rate of upper respiratory tract infection was similar in the BAFF or APRIL-targeted drug and placebo groups (RR 2.02; 95% CI 0.37 to 11.15; $p = 0.4208$; $I^2=51.7\%$; Fig. 4B). In the sensitivity analysis using the SJHK method, the outcome was consistently maintained (RR 1.69; 95% CI 0.43 to 6.63; $p = 0.4538$; $I^2=33.3\%$; Figure S4A). In the leave-one-out sensitivity analysis, the removal of each study from the pooled analysis did not affect this outcome, and the removal of the Jicheng Lv 2023 trial removed heterogeneity ($I^2=0\%$; Figure S4B).

Specifically evaluating the presence of drug or placebo-related adverse events, there was no significant difference between the groups (RR 1.30; 95% CI 0.87 to 1.93; $p = 0.2001$; $I^2=0\%$; Figure S5A). In the sensitivity analysis, the outcome was maintained consistently using the SJHK method (RR 1.37; 95% CI 0.73 to 2.60; $p = 0.3286$; $I^2=0\%$; Figure S5B) and in the leave-one-out analysis (Figure S5C).

Regarding the presence of serious adverse effects, there was no significant difference between the BAFF or APRIL-targeted drug and placebo groups (RR 0.74; 95% CI 0.30 to 1.86; $p = 0.5215$; $I^2=0\%$; Figure S6A). In the sensitivity analysis, the outcome remained significant using the SJHK method (RR 0.75; 95% CI 0.27 to 2.07; $p = 0.5793$; $I^2=0\%$; Figure S6B) and in the leave-one-out analysis (Figure S6C).

Regarding the rate of adverse events that led to discontinuation of the study, no significant differences were found between the BAFF or APRIL-targeted drug and placebo groups (RR 0.79; 95% CI 0.14 to 4.93; $p = 0.7893$; $I^2=0\%$; Figure S7A). In the sensitivity analysis, the outcome was consistently maintained using the SJHK method (RR 0.79; 95% CI 0.14 to 4.48; $p = 0.7924$; $I^2=0\%$; Figure S7B) and in the leave-one-out analysis (Figure S7C).

Biomarkers

The mean percent change in serum Gd-IgA1 from baseline showed a significant decrease with the use of BAFF or APRIL-targeted drugs compared to placebo (MD -46.48%; 95% CI -63.46 to -29.49; $p < 0.0001$; $I^2=61.2\%$; Fig. 5). In the sensitivity analysis using the SJHK method, the superior effect of BAFF or APRIL-targeted drugs over placebo was maintained (MD -46.43%; 95% CI -60.77 to -32.08; $p < 0.0001$; $I^2=61.2\%$; Figure S8A). In the leave-one-out sensitivity analysis, the removal of each study from the pooled analysis did not affect this endpoint, and homogeneity was achieved by omitting ORIGIN 2024 or ENVISION 2023 (Figure S8B).

About the serum IgG level, the mean percent change from baseline showed a significant decrease with the use of BAFF or APRIL-targeted drugs compared to placebo (MD -29.76%; 95% CI -34.48 to -25.03; $p < 0.0001$; $I^2=0\%$; Figure S9A). In the sensitivity analysis using the SJHK method, the superior effect of BAFF

or APRIL-targeted drugs over placebo was maintained (MD -29.65%; 95% CI -35.89 to -23.42; $p < 0.0001$; $I^2=0\%$; Figure S9B). Specifically, the serum IgG level (g/L) at the end of the study was significantly lower in the BAFF or APRIL-targeted drug group compared to placebo (MD -3.11 g/L; 95% CI -4.00 to -2.22; $p < 0.0001$; $I^2=0\%$; Figure S9C). In the sensitivity analysis, the outcome remained consistent using the SJHK method (MD -3.00 g/L; 95% CI -4.13 to -1.86; $p < 0.0001$; $I^2=0\%$; Figure S9D) and in the leave-one-out analysis (Figure S9E).

Concerning the serum IgA level, the mean percent change from baseline showed a significant decrease with the use of BAFF or APRIL-targeted drugs compared to placebo (MD -49.04%; 95% CI -57.53 to -40.54; $p < 0.0001$; $I^2=0\%$; Figure S10A). In the sensitivity analysis using the SJHK method, the superior effect of BAFF or APRIL-targeted drugs over placebo was maintained (MD -49.04%; 95% CI -57.53 to -40.54; $p < 0.0001$; $I^2=0\%$; Figure S10B). Specifically, the serum IgA level (g/L) at the end of the study was significantly lower in the BAFF or APRIL-targeted drug group (MD -1.20 g/L; 95% CI -2.07 to -0.32; $p = 0.0072$; $I^2=0\%$; Figure S10C). In the sensitivity analysis, the outcome did not change (MD -1.20 g/L; 95% CI -2.07 to -0.32; $p = 0.0072$; $I^2=0\%$; Figure S10D).

In terms of serum IgM level, the mean percent change from baseline showed a significant decrease with the use of BAFF or APRIL-targeted drugs compared to placebo (MD -65.56%; 95% CI -75.62 to -55.49; $p < 0.0001$; $I^2=0\%$; Figure S11A). In the sensitivity analysis using the SJHK method, the superior effect of BAFF or APRIL-targeted drugs over placebo was maintained (MD -65.55%; 95% CI -75.64 to -55.47; $p < 0.0001$; $I^2=0\%$; Figure S11B). Specifically, the serum IgM (g/L) level at the end of the study was significantly lower in the BAFF or APRIL-targeted drug group compared to placebo (MD -0.59 g/L; 95% CI -0.77 to -0.40; $p < 0.0001$; $I^2=0\%$; Figure S11C). It did not change with the sensitivity analysis, (MD -0.59 g/L; 95% CI -0.85 to -0.34; $p < 0.0001$; $I^2=0\%$; Figure S11D).

DISCUSSION

This systematic review and meta-analysis analyzed the efficacy of BAFF and APRIL inhibitors in adult patients with biopsy-confirmed IgAN. We found four similarly designed RCTs that compared these therapies with placebo. We consistently showed that BAFF or APRIL inhibitors significantly decreased 24-hour UPCR and increased eGFR compared to placebo; the drugs were safe and tolerable for patients 18 years or older. In addition, BAFF or APRIL inhibitors significantly decreased serum levels of Gd-IgA1, IgG, IgA, and IgM compared to placebo.

Regarding the mean change in 24-hour UPCR from baseline, our study found a significant reduction in proteinuria that favors BAFF and APRIL inhibitors over placebo. Although the pathophysiology of IgAN is complex, robust evidence suggests that sustained stabilization of eGFR and proteinuria may halt the progression to ESRD [19–21]. To date, therapies targeting Gd-IgA1 and its autoantibody stand out as some of the most promising treatments for IgAN. While sibeprenlimab exclusively inhibits APRIL signaling; atacicept and telitacicept are dual APRIL/BAFF antagonists. Although these two cytokines share considerable homology, their specific interactions with receptors are of particular interest in IgAN

studies [22]. BAFF uniquely binds to the BAFF receptor (BAFF-R), mainly expressed on immature and mature naïve B cells. However, both BAFF and APRIL interact with B-cell maturation antigen (BCMA), transmembrane activator, calcium modulator, and cyclophilin ligand interactor (TACI) [23].

These findings align with other immunological therapies for IgAN. Zigakibart, a humanized IgG4 monoclonal antibody targeting APRIL, has shown promising interim results in a phase 2 trial for IgAN, with substantial reductions in proteinuria and a significant decrease in serum IgA and Gd-IgA1 levels [23–25]. For this reason, a phase 3 trial (ClinicalTrials.gov number NCT05852938) [26] is underway to further assess its efficacy and safety in slowing the progression of renal dysfunction in IgAN patients. In parallel, a targeted release formulation of budesonide (TRF-budesonide) also led to a reduction in proteinuria in patients with IgAN by downregulating aberrant IgA production [27]. In addition, several other therapies, including complement pathway inhibitors, protein degradation inhibitors, and endothelin receptor antagonists, are under investigation in clinical trials. The findings will help us better understand the pathophysiology of IgAN and allow comparisons between these novel immune therapies [28].

Safety was evaluated based on the rates of any adverse events, drug- or placebo-related adverse events, serious adverse events, adverse events leading to discontinuation and upper respiratory tract infection. The incidence was similar in the treated groups compared to placebo. As expected, all studies showed reductions in Gd-IgA1, IgA, IgG, and IgM levels in patients treated with atacicept, telitacicept, or sibeprenlimab, which was not observed in the placebo group [5–8]. However, no increase in infection-related adverse events was observed. Importantly, the trials analyzed were conducted during the COVID-19 pandemic [5–8]. Although the authors cited the limited number of participants as a study limitation, the absence of infection-related adverse events may be attributed to selective inhibition of APRIL and BAFF, thus avoiding significant lymphocyte depletion [29]. However, long-term safety studies are necessary to assess the risk of immunosuppression, as patients will likely require repeated dosing to obtain sustained responses.

Our meta-analysis supports the idea that targeting the APRIL/BAFF signaling pathway in IgAN may be promising in terms of efficacy and safety. However, certain limitations in our data deserve careful consideration. The few available phase II studies, along with slight variations in design and small sample sizes, prevent the possibility of conducting in-depth subgroup analyses and yield a moderate quality of evidence. The ENVISION (2023) trial [7] showed some baseline differences in the percentage of crescents and the time since the biopsy, which could affect the efficacy results to some extent. In the study by Jicheng Lv et al. (2023) [8], the short follow-up period limited the evaluation of the long-term efficacy of telitacicept on kidney function. Furthermore, there was an imbalance of confounders between the three analyzed groups (placebo, telitacicept 160 mg and telitacicept 240 mg), with a higher proportion of women, patients with lower body mass index, and more people with a history of immunosuppressive therapy in the high-dose groups, resulting in a higher dose of telitacicept per kilogram. Additionally, the slightly accelerated decrease in eGFR in the placebo group may be attributed to the small sample size. In the ORIGIN (2024) trial [5], limitations included a disproportionate number of white participants assigned to the placebo group compared to those receiving atacicept, which may

affect the generalizability of the findings to other populations. Additional studies are also needed to assess the effect of different types and doses of BAFF/APRIL inhibitors. It should be mentioned that the efficacy and safety of sibeprenlimab in a larger population of patients with IgAN are being investigated in an ongoing phase 3 trial (ClinicalTrials.gov number NCT05248646) [30].

To our knowledge, this is the first systematic review and meta-analysis that evaluates the efficacy and safety of drugs targeting BAFF or APRIL in adults with IgAN. Taken together, our results indicate that BAFF and APRIL-targeted drugs appear to be clinically effective and safe for the improvement of proteinuria, eGFR, and serum biomarkers in patients with IgAN. Larger, phase III RCTs with comprehensive subgroup analyses and extended long-term data are required to support our findings and thoroughly assess the clinical feasibility of BAFF- or APRIL-targeted drugs in IgAN.

Declarations

Conflict of interest: All authors declare that they have no affiliations or involvement with any organization or entity with a financial or nonfinancial interest in the subject matter or materials addressed in this manuscript.

Ethical approval: This is not applicable as the manuscript is a review paper.

Human and animal rights: This article does not include any studies involving human participants or animals conducted by the authors.

Informed consent: This is not applicable as the manuscript is a review paper.

Data availability: The primary data that support the findings of this study are derived from published literature, and the relevant citations are provided in the references section. The raw data sets analyzed and the collection forms are available from the corresponding author upon reasonable request.

CRedit Author Contributions: **Conceptualization:** R.d.S.B., A.C.S.e.S.; **Data curation:** L.H.d.P., R.d.S.B.; **Formal analysis:** R.d.S.B., P.A.M.G.d.S.; **Investigation:** L.H.d.P., R.d.S.B.; **Methodology:** R.d.S.B., J.M.B.M.; **Project administration:** R.d.S.B., A.C.S.e.S.; **Supervision:** R.d.S.B., A.C.S.e.S.; **Validation:** G.G.B., J.M.B.M.; **Visualization:** R.d.S.B., L.H.d.P., R.L.S., P.A.M.G.d.S.; **Writing – original draft:** R.d.S.B., J.M.B.M., G.G.B., R.L.S., L.H.d.P., P.A.M.G.d.S.; **Writing – review & editing:** A.C.S.e.S., R.d.S.B., G.G.B.

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Table 1

Table 1 is available in the supplemental file section.

Figures

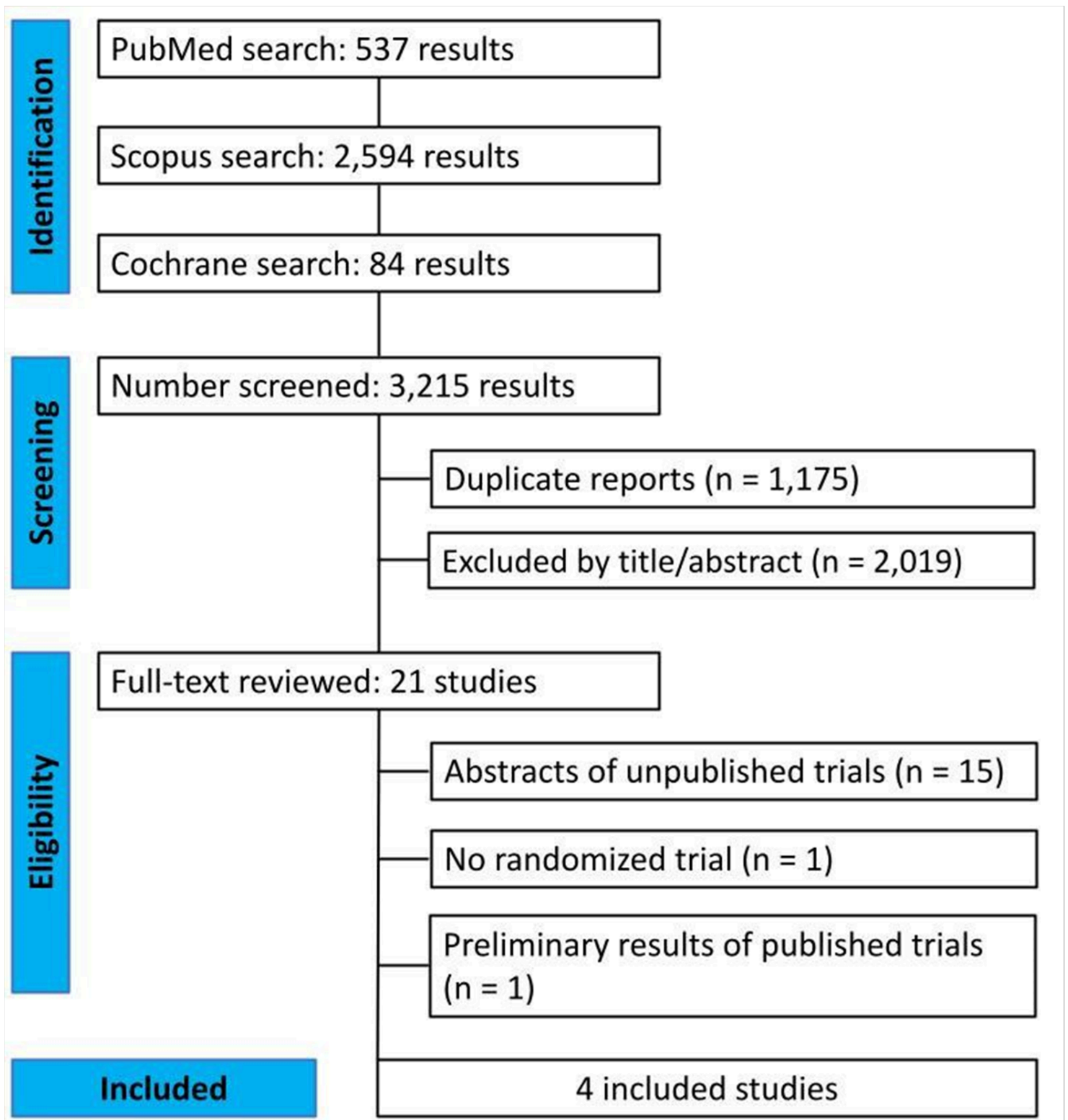


Figure 1

PRISMA flow diagram of study screening and selection

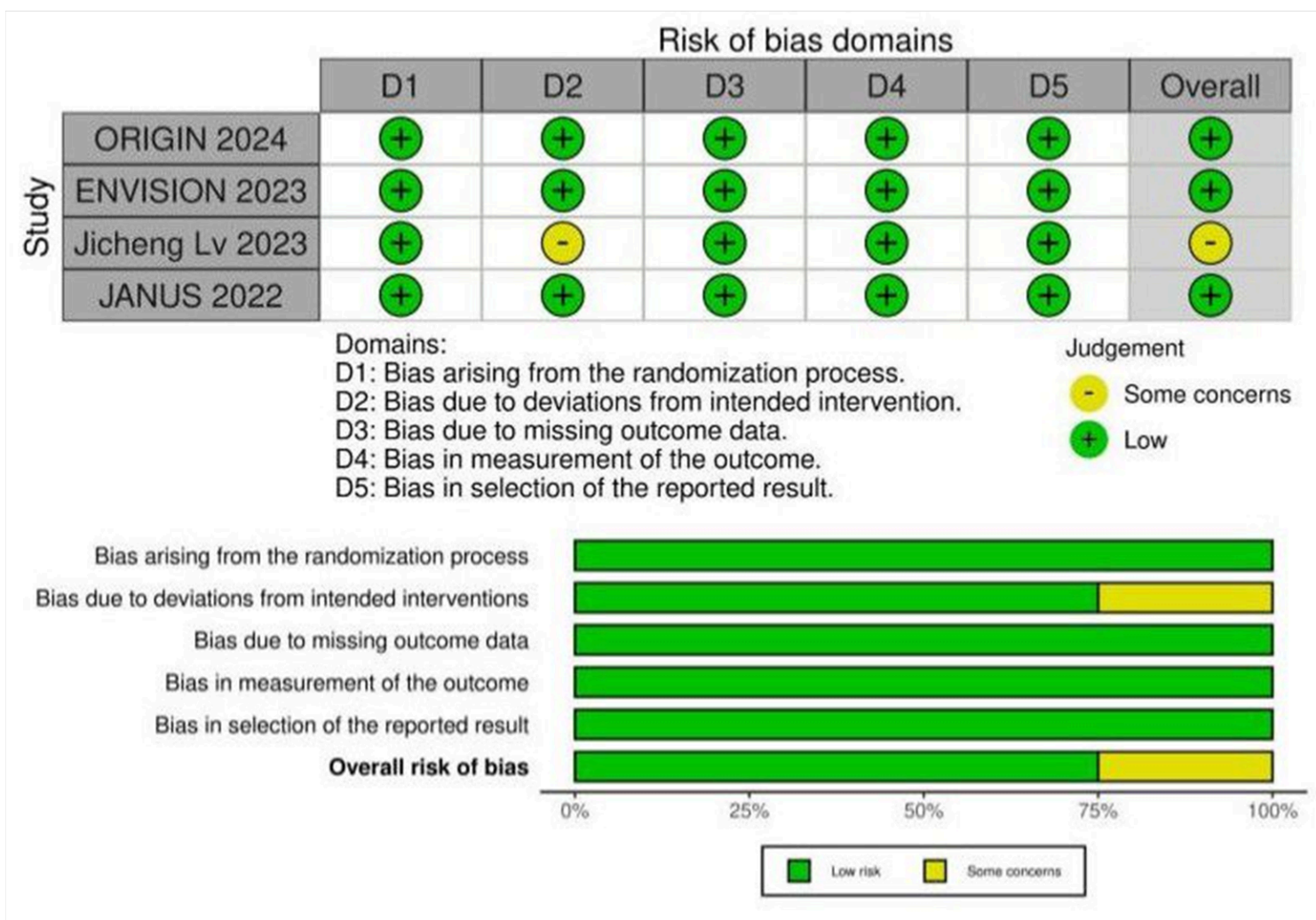


Figure 2

RoB2 Risk of bias assessment

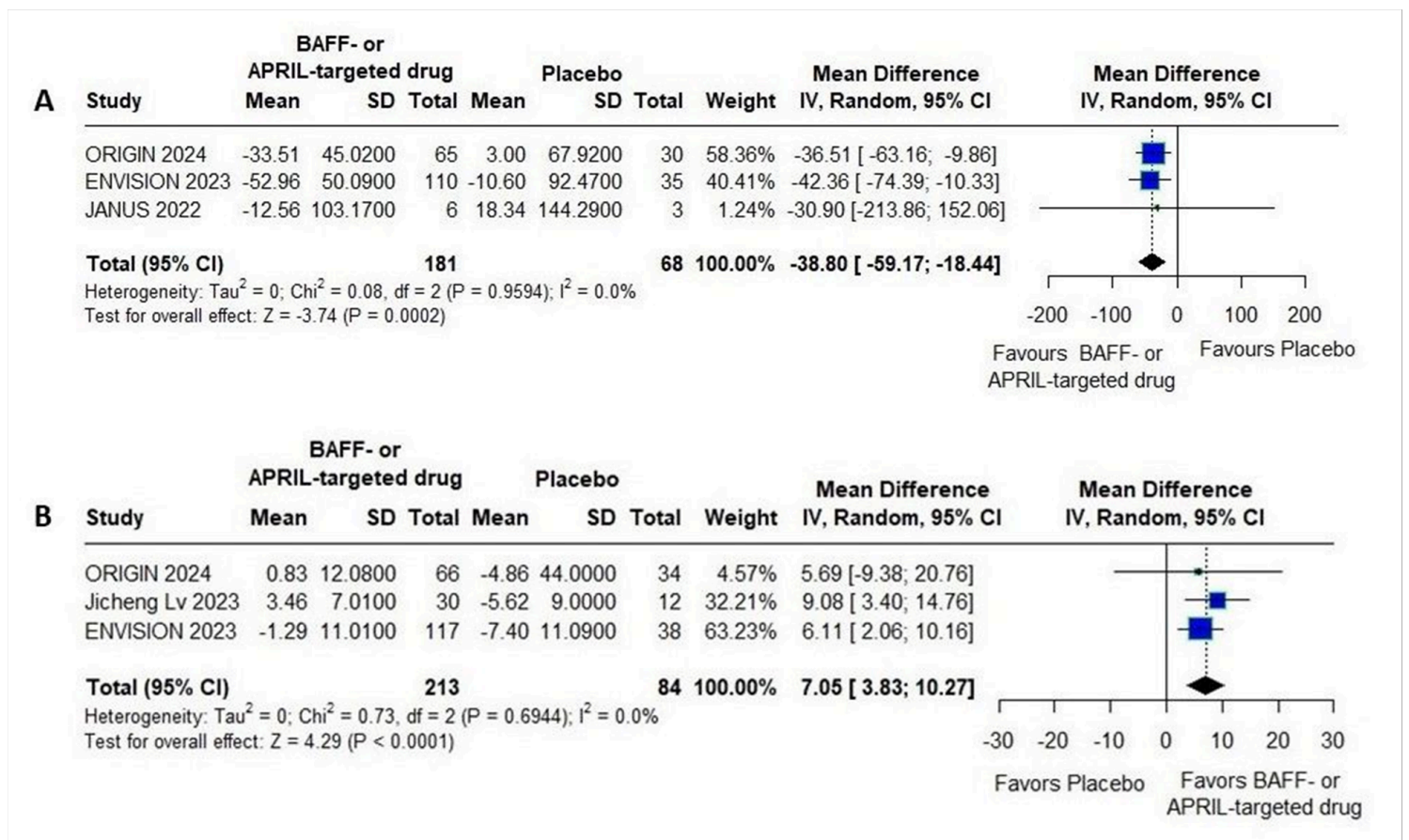


Figure 3

Efficacy of the BAFF- or APRIL-targeted drug. **(A)** Meta-analysis of the mean percent change in UPCR for 24 hours from baseline, **(B)** Meta-analysis of the mean change in eGFR from baseline

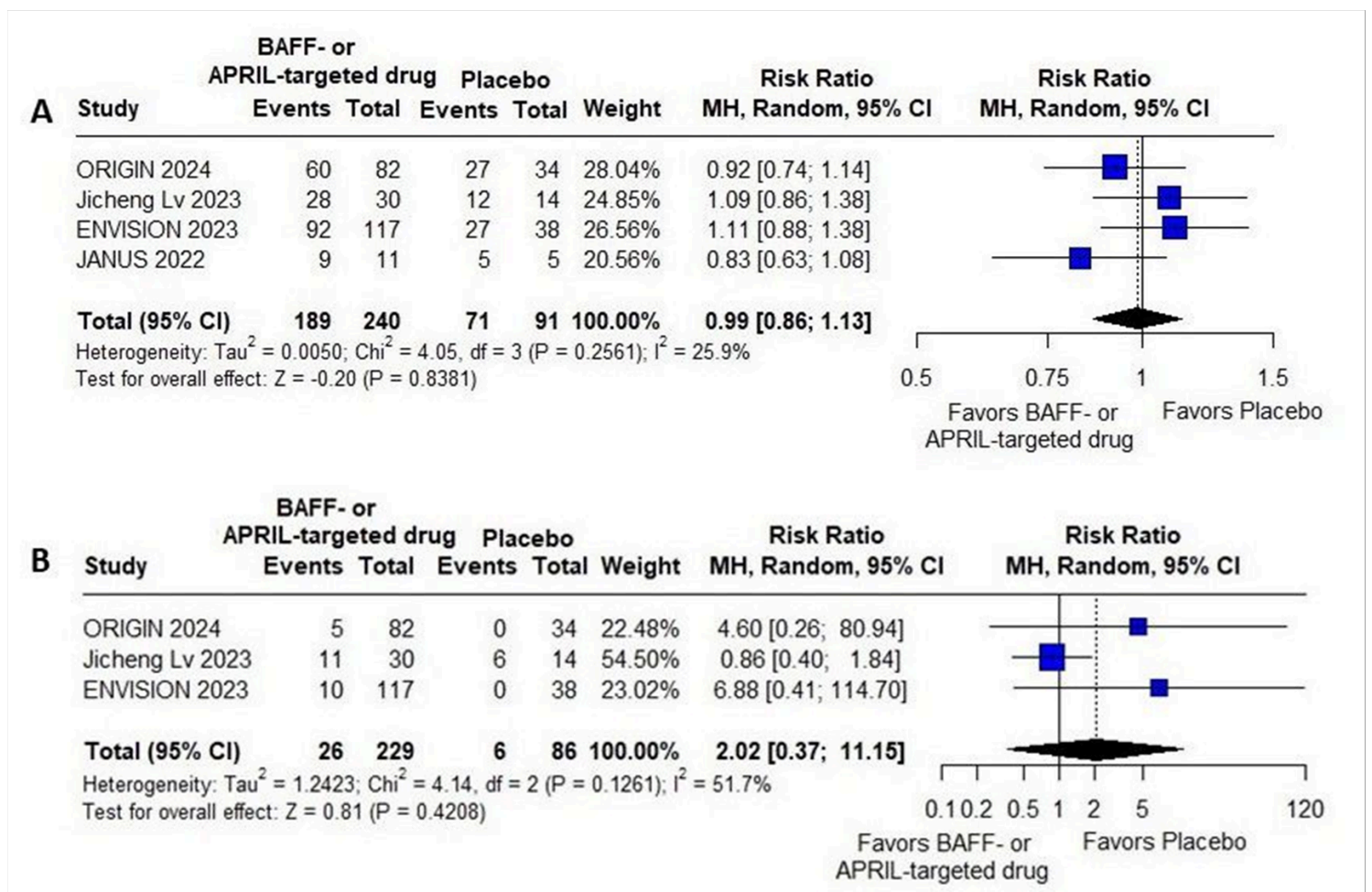


Figure 4

Safety of the BAFF or APRIL-targeted drug. **(A)** Meta-analysis of the rate of any adverse effect; **(B)** Meta-analysis of the rate of upper respiratory tract infection

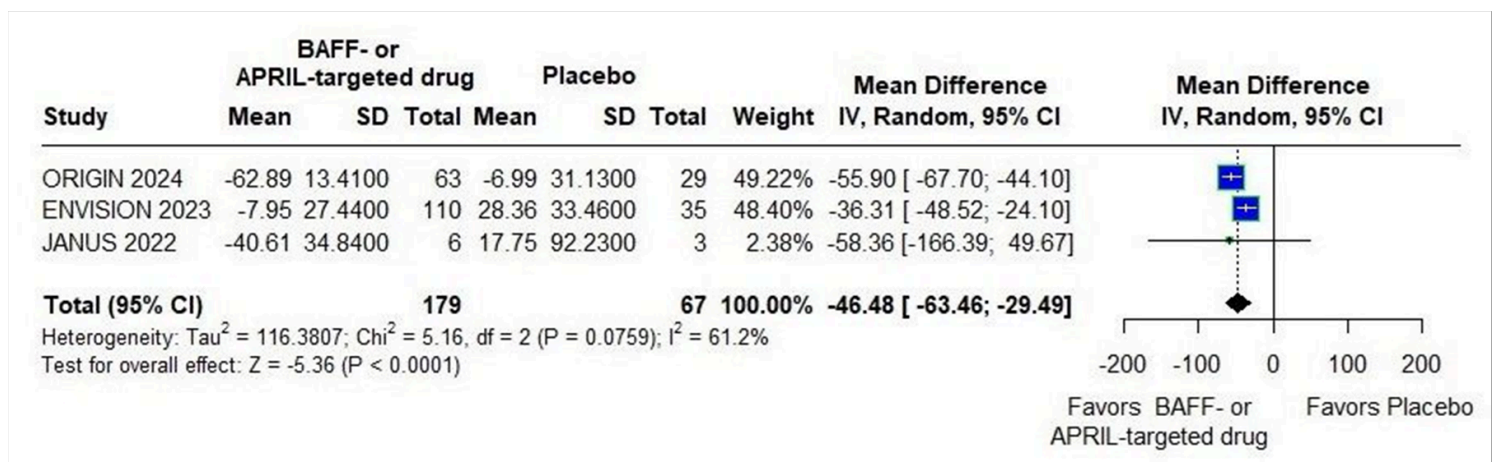


Figure 5

Meta-analysis of the mean percentage change in serum Gd-IgA1 from baseline

Supplementary Files

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- [Table1.docx](#)
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